

MARILYN, Antarctica

WMO: 898690

Lat: 79.904S

Lon: 165.774E

Elev: 204

StdP: 14.59

Time Zone: 12.00 (NZL)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-60.4	-55.1	N/A	N/A	N/A	N/A	N/A	N/A	42.9	-20.0	38.7	-18.4	5.4	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.9	33.7	N/A	30.7	N/A	28.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.3	31.4	26.9	DB	-67.0	43.5	5.1	4.3	-70.6	46.6	-73.6	49.1	-76.4	51.5	-80.1	54.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.9	20.6	3.6	-13.2	-19.2	-22.1	-21.9	-25.4	-26.0	-20.9	-10.7	7.0	21.9	(6)	
	DBStd	20.08	5.97	9.03	10.02	11.54	13.87	12.66	13.05	13.46	12.74	8.94	9.13	5.10	(7)	
	HDD50	21502	913	1300	1958	2076	2235	2158	2337	2357	2126	1881	1289	872	(8)	
	HDD65	26977	1378	1720	2423	2526	2700	2608	2802	2822	2576	2346	1739	1337	(9)	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	11.1	6.3	8.0	11.9	13.6	13.1	13.0	14.7	12.5	13.9	12.1	8.0	5.7	(14)	
	PrecAvg	6.9	0.7	0.7	0.6	0.5	0.6	0.6	0.5	0.6	0.6	0.5	0.6	0.6	(15)	
(16) Precipitation	PrecMax	11.1	1.7	1.5	1.6	1.5	1.3	1.6	1.5	2.1	1.3	1.4	1.4	1.2	(16)	
	PrecMin	2.3	0.1	0.2	0.1	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.0	(17)	
	PrecStd	1.9	0.4	0.4	0.3	0.3	0.3	0.4	0.3	0.5	0.4	0.4	0.3	0.3	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	27.7	15.2	16.2	13.3	12.6	10.3	10.1	15.0	14.9	27.4	41.0	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	34.0	21.9	9.3	6.5	7.2	6.1	4.2	3.5	7.5	10.8	24.7	34.8	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	30.9	18.8	5.5	1.7	1.5	0.6	-2.0	-0.9	0.9	7.6	21.9	31.4	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	28.7	16.4	2.0	-2.6	-3.6	-4.2	-7.6	-6.5	-4.3	3.3	19.3	29.3	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(36) Mean Daily Temperature Range		MDBR	13.5	14.5	14.3	14.1	15.6	15.7	14.7	16.7	14.9	15.4	14.8	13.9	(35)	
	5% DB	MCDBR	16.7	15.9	14.7	18.5	20.0	18.1	16.9	18.8	16.4	18.9	15.5	17.6	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub	0.230	0.208	0.162	N/A	N/A	N/A	N/A	N/A	N/A	0.153	0.190	0.211	0.228	(40)	
	taud	2.183	2.189	2.094	N/A	N/A	N/A	N/A	N/A	N/A	2.052	2.179	2.199	2.192	(41)	
	Ebn at Noon	302	280	227	0	0	0	0	0	0	226	288	310	312	(42)	
	Edh at Noon	31	24	13	0	0	0	0	0	0	13	24	31	34	(43)	
(44) All-Sky Solar Radiation	RadAvg	2599	1524	445	0	0	0	0	0	0	210	1108	2433	3031	(44)	
	RadStd	80	65	23	0	0	0	0	0	0	12	39	76	76	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page